AVIRAL GUPTA

SOFTWARE ENGINEER 2 · TEST INFRASTRUCTURE & DEVELOPER PRODUCTIVITY

guptaavi352@gmail.com • +91 7726924322 • India • linkedin.com/in/aviral-gupta-754341152

SUMMARY

SDE 2 on Postman's Test Infrastructure and Developer Productivity team. Over the past 3+ years I've built the CI/CD, performance, and AI tooling that hundreds of engineers rely on to ship a the Postman web and electron desktop app quickly and safely. My work has spanned E2E automation, performance guardrails, coverage governance, and AI-assisted test triage. I like improving systems that are slow and manual and making them fast and self-serve.

RECOGNITION

> *"You have operated like a one-man army across multiple initiatives, embracing emerging AI technologies and translating them into practical, meaningful offerings for the team — and your work has been central to shifting the narrative around Test Infra, which is no longer viewed as a blocker to feature releases."*
>
> — **Salil Gupta, Engineering Manager, Postman**

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EXPERIENCE

Postman — Software Engineer 2

Jan 2023 – Present

Test Infrastructure & Developer Productivity · India

E2E Test Platform & CI/CD

- Migrated the slow, flaky WebdriverIO suite to Playwright. WDIO had piled up wrappers and a CJS teleport framework that transpiled our ES specs, which made builds slow, and its reports needed a script just to attach logs and had no timeline view. Playwright brought native ES, much easier debugging, CDP execution (~2x faster than WDIO's ChromeDriver hops), and agent-scaffolded specs that **cut new-spec authoring ~80% (~1 story point → ~30–60 min)**.
- Designed the framework to be modular and scalable, keeping the core offerings (hydration, selective test execution, ticket creation, and performance checks) in a standalone automations package that any project can drop in through an agent.
- Built change-based selective execution so a PR runs only the specs that exercise its changed code. E2E specs don't import source directly, so the mapping is derived from coverage: each run's V8 coverage gives the compiled chunk names, which resolve through source-maps back to the executed source lines, producing a reverse source-to-spec map that selects specs when their files change. This **catches ~90% of leaked mainline bugs on the PR itself**.
- Fixed flaky Windows E2E runs caused by a Cloudflare allowlist race condition: an AWS Lambda and SQS pipeline registers each runner's dynamic IP before its run and clears it afterward.

AI & Agentic Test Tooling

- Built and deployed the test-infra-mcp-server, adding tools that let developers trigger E2E tests and check their squad's flaky specs, plus tools that let agents automatically triage consistent E2E failures. Roughly 63% of triage tickets were resolved without manual intervention.
- Designed E2E Coverage Signals, a dashboard that maps where customers actually feel pain (from GitHub, customer escalations, incident.io, and Slack) against our E2E coverage and ranks the riskiest gaps with a transparent scoring formula. A daily AI agent feeds the deterministic Node pipeline behind it, shipped as a React/Vite app on GitHub Pages. It gave leadership their first clear read on which important areas are under-tested.

Coverage & Quality Governance

- Shipped an incremental-coverage check as a reusable GitHub Action across every backend repo. Instead of reporting where total coverage sits, it tells the author whether the lines they just wrote are tested. Company-wide unit and integration coverage increased from **42% to 73% in 3 months**.

Performance & Release Guardrails

- Created a release-confidence score that **holds back a tier rollout until it clears 95%**. Automated daily-release analysis and summaries took over most of the **~4–5 hr manual signoff**, a big part of how Postman now ships **daily, in under an hour**.
- Measured app warm-boot times through a Playwright → CircleCI → New Relic pipeline and turned them into release gates. Built the self-serve infra behind it so developers can trigger runs on demand across every OS, see how their PR affects performance, and compare it against earlier builds.

Operations

- Moved on-call alerting from Opsgenie to incident.io — schedules, routing, and a shared helper package.
- Built squad and company dashboards for incident, bug, and CUE trends. They feed the State of Quality dashboard in Looker, which gives the org a running view of reliability.
- **Cut E2E test setup time from ~30s to ~6s** by creating users in parallel, moving user cleanup to async monitors, removing barrel-file imports, and caching repeated network calls.

SKILLS

AI & Agentic tooling: MCP servers, LLM agents, Cursor/Claude Automations

Test automation: Playwright, WebdriverIO, Puppeteer, Allure reporting, NYC/Istanbul, page-object design

CI/CD & build: CircleCI, GitHub Actions, Nx monorepo, Yarn v4, Selective Test Execution

Observability & data: New Relic (NRQL), Looker, AWS (Lambda, SQS, Redshift, S3), SonarQube, incident.io, Opsgenie

Performance: App relaunch benchmarking + regression gating, critical-path bundle-size gates, Lighthouse Metrics (LCP, FID, CLS)

Languages: JavaScript, C++

EDUCATION

Indian Institute of Information Technology, Allahabad — B.Tech, Information Technology

Jul 2019 – Jul 2023

CGPA 8.53